# Regression Diagnostics

POLS 602

Fall 2026

Mike Burnham

Texas A&M University

# Review

1. What does LOESS let the data decide that a quadratic regression does not?
2. What does quantile regression estimate that OLS does not?
3. What are the Gauss-Markov assumptions?

# Today

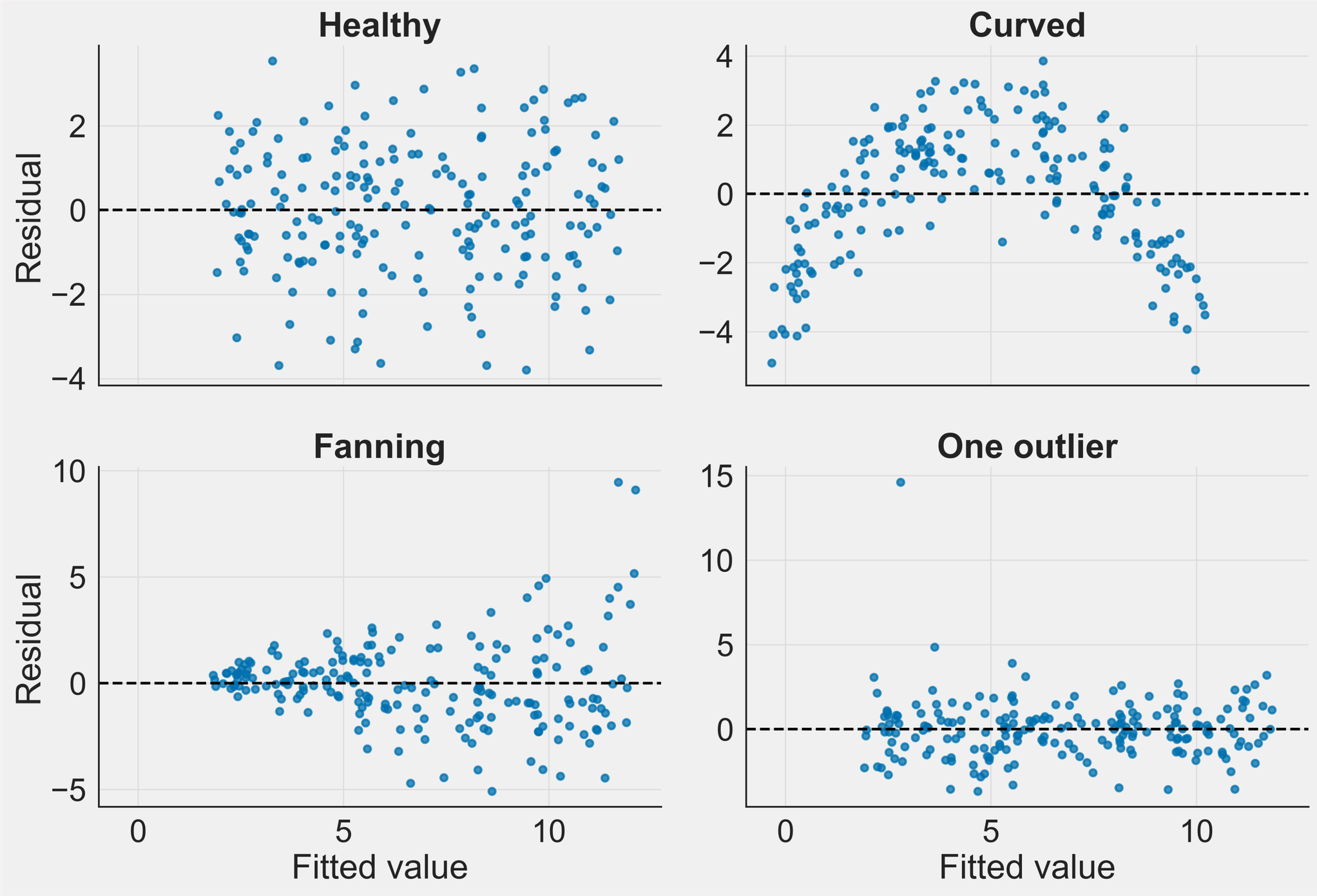
- How do we check whether the assumptions hold?
- What breaks when the errors are not normal?
- What breaks when the errors do not have constant variance?
- What breaks when the errors are correlated within groups?

# Checking the Model

# Assumptions Are About Errors

- The assumptions are about the errors $\varepsilon_i$
- We never observe the errors
- We observe the residuals $\hat{\varepsilon}_i = y_i - \hat{y}_i$
- Diagnostics look for structure in the residuals

# Reading a Residual Plot



# Three Problems, Three Questions

| Problem | What to check |
|---|---|
| Non-normal residuals | Is the shape bell-like? |
| Heteroscedasticity | Is the spread constant? |
| Clustered errors | Do the errors move together within groups? |

# Non-Normal Residuals

# The Problem

- Normality assumes $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$
- Residuals may be skewed, heavy-tailed, or lumpy
- *Do the residuals look like a draw from a bell curve?*

# Why It Matters

- Unbiasedness and BLUE do not need normality
- Exact $t$ and $F$ tests do
- In large samples the CLT makes the tests approximately right
- If you want intervals on a single prediction though, the CLT doesn't help you

# What Causes It

- A skewed outcome, such as income, wages, or counts
- Nonlinearity or omitted variables that the model leaves in the error
- Outliers or data errors
- Mixed groups with different means
- Bounded or binary outcomes

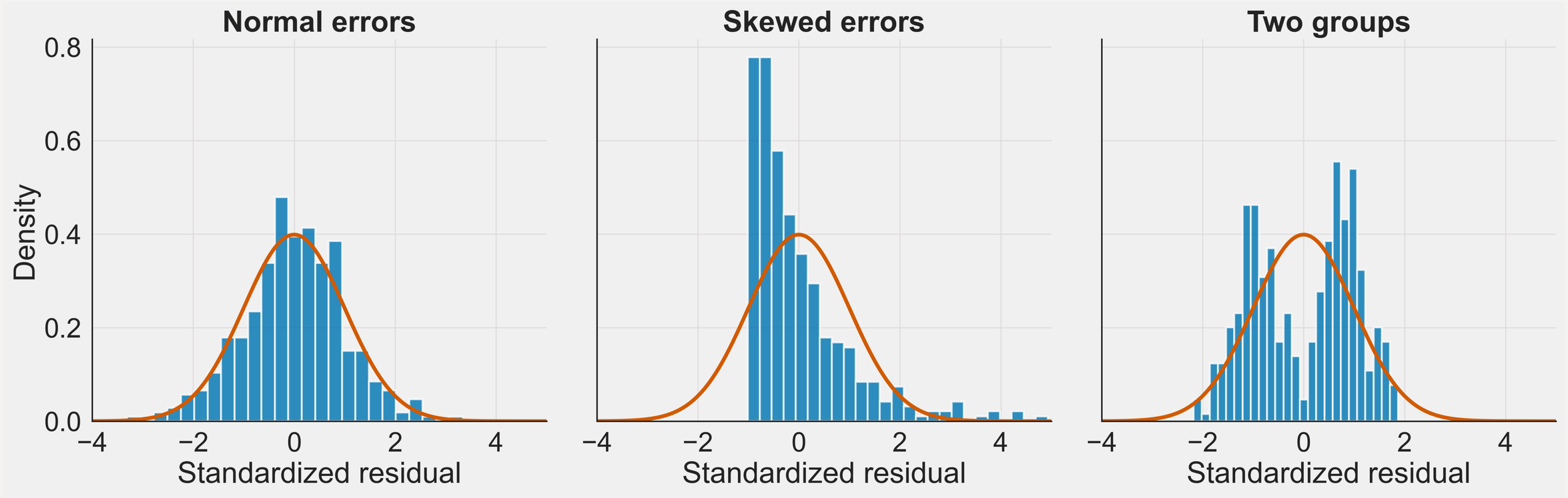
# What It Implies

- Non-normality is usually a symptom, not the disease
- Check the functional form and the outcome’s scale first
- Check for outliers and mixed groups second
- Do not fix the residuals without asking why they look this way

# Residual Histogram

- Plot a histogram of the residuals
- Overlay a normal curve with the same mean and variance
- Look at the shape: skew, tails, humps

# Histograms Show Skew and Humps



# What Is a Quantile?

- **Quantile**: the value $q_\tau$ below which a share $\tau$ of the distribution falls
- The 0.5 quantile is the median, and 0.25 and 0.75 are the quartiles
- Percentiles are quantiles written as percents
- *What value has 97.5% of a standard normal distribution below it?*

$$P(Z \leq q_\tau) = \tau, \qquad q_{0.975} = 1.96$$

# What Is a QQ Plot?

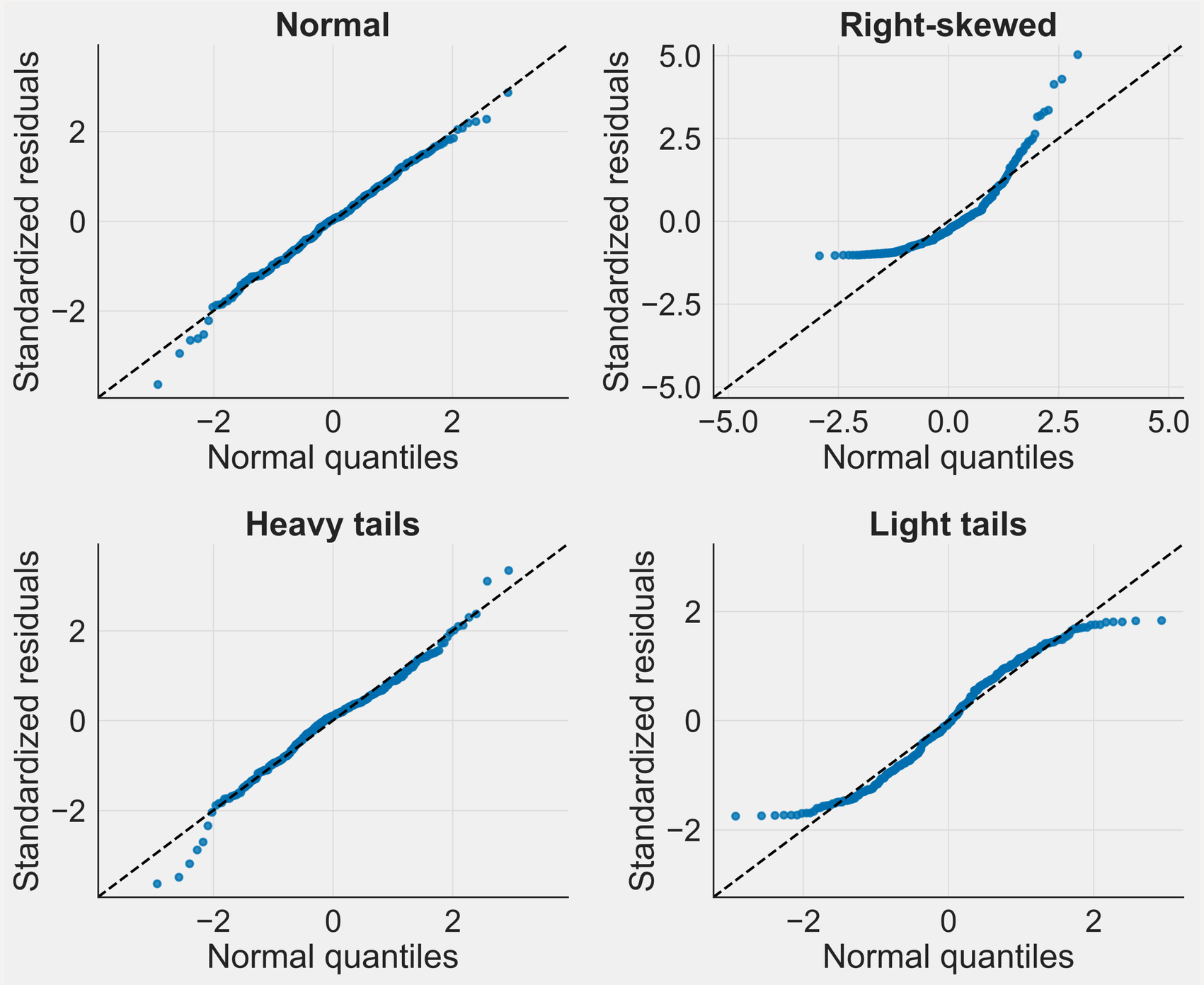
- **QQ plot** (quantile-quantile plot): the quantiles of the residuals plotted against the quantiles of a normal distribution
- If the residuals are normal, the points fall on a straight line
- Where the points leave the line shows how the shape differs from normal

- *Do the residuals have the quantiles that a normal distribution would give?*

# How a QQ Plot Is Built

1. Standardize the residuals and sort them from smallest to largest
2. For each rank $i$, find the normal quantile at $(i - 0.5)/n$
3. Plot the sorted residuals against the normal quantiles
4. Add the 45-degree line

# Reading a QQ Plot



# Formal Tests of Normality

- A formal test asks whether the residuals are consistent with a normal distribution
- Null hypothesis: the residuals are normal
- A small $p$-value is evidence against normality
- Two common tests: Shapiro-Wilk and Jarque-Bera

# The Shapiro-Wilk Test

$$W = \frac{\left( \sum_{i=1}^{n} a_i \, \hat{\varepsilon}_{(i)} \right)^2}{\sum_{i=1}^{n} \hat{\varepsilon}_i^{\,2}}$$

- $\hat{\varepsilon}_{(i)}$: the $i$th smallest residual
- $a_i$: weights set by what a normal sample's ordered values look like
- $W$ lies between 0 and 1, and $W$ near 1 means normal
- Small $W$ rejects normality

# The Jarque-Bera Test

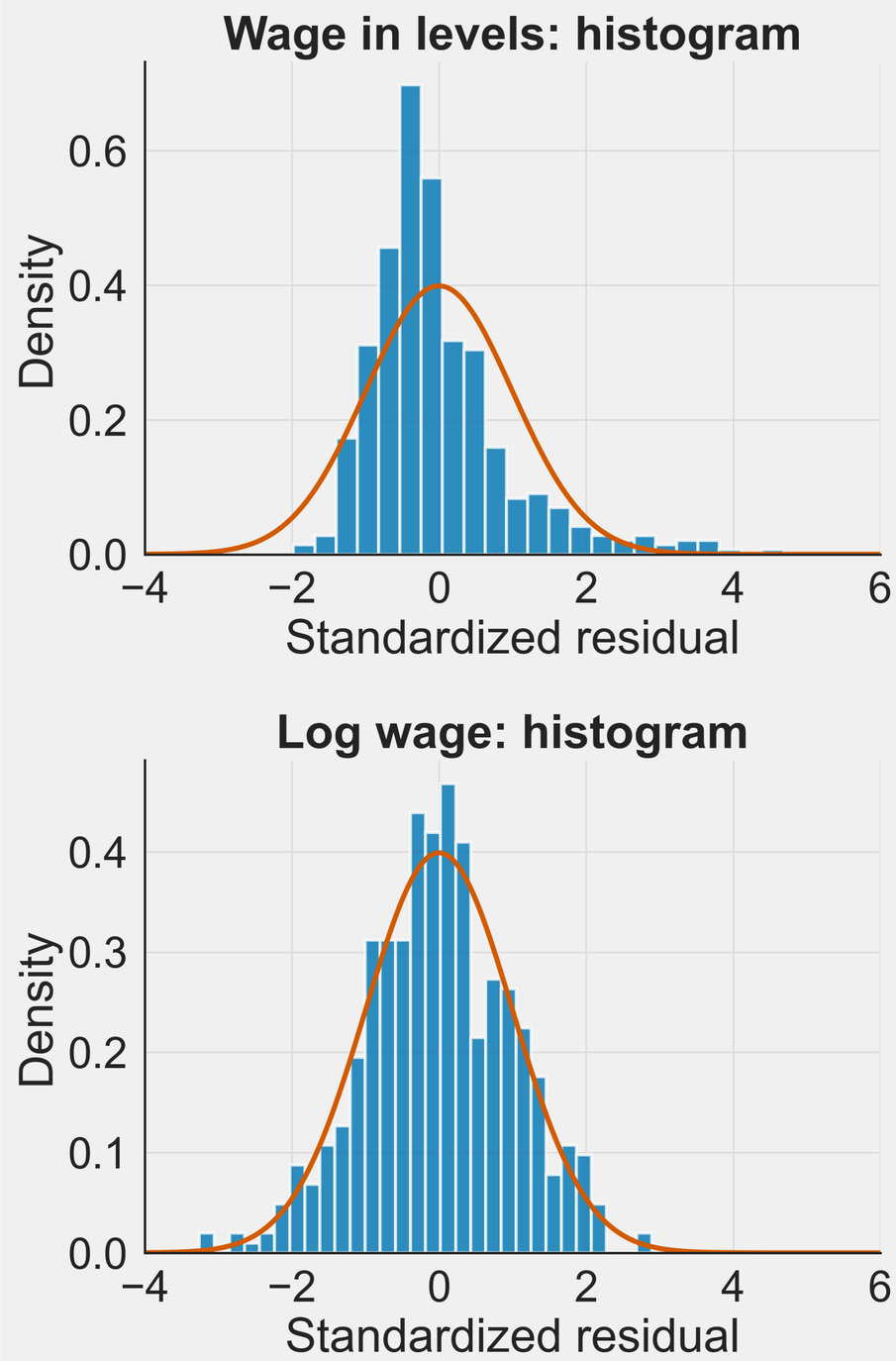
$$JB = \frac{n}{6}\left(g_1^2 + \frac{g_2^2}{4}\right)$$

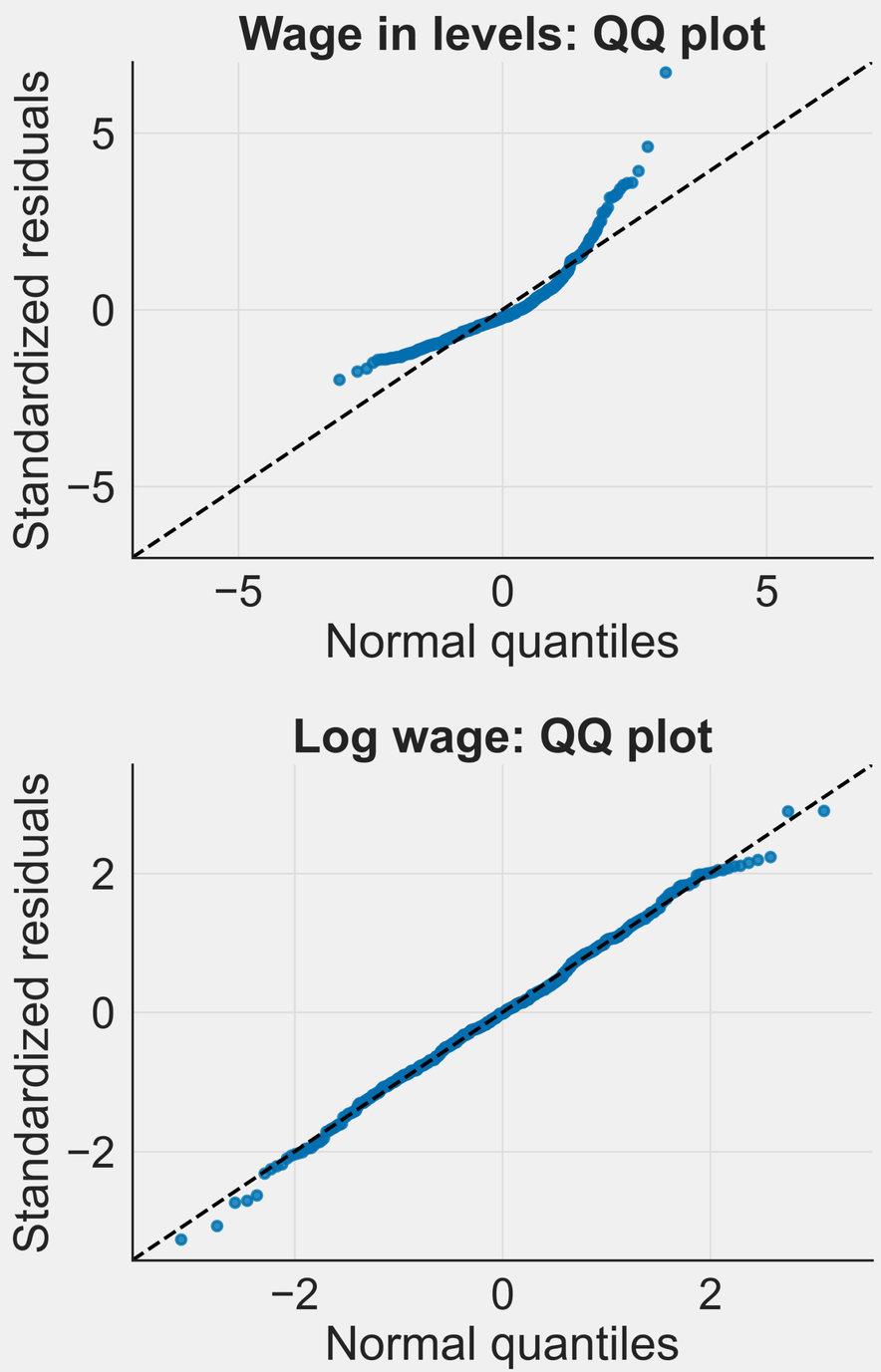
- $JB$ is large when skewness ($g_1$) or excess kurtosis ($g_2$) is far from 0
- Reject at 5% if $JB > 5.99$

# Cautions With Normality Tests

- Large samples reject for departures too small to matter
- Small samples have little power to detect real departures
- Failing to reject does not show the residuals are normal
- Use the tests alongside the plots, never in place of them

# Skewed Outcome, Skewed Residuals



# What to Do

- Find the cause before choosing a fix
- Transform a skewed outcome (logs)
- Repair the functional form or add missing groups
- Investigate outliers
- With a large sample and a focus on coefficients, rely on the CLT

# Heteroscedasticity

# The Problem

$$\text{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma_i^2 \quad \text{instead of} \quad \sigma^2$$

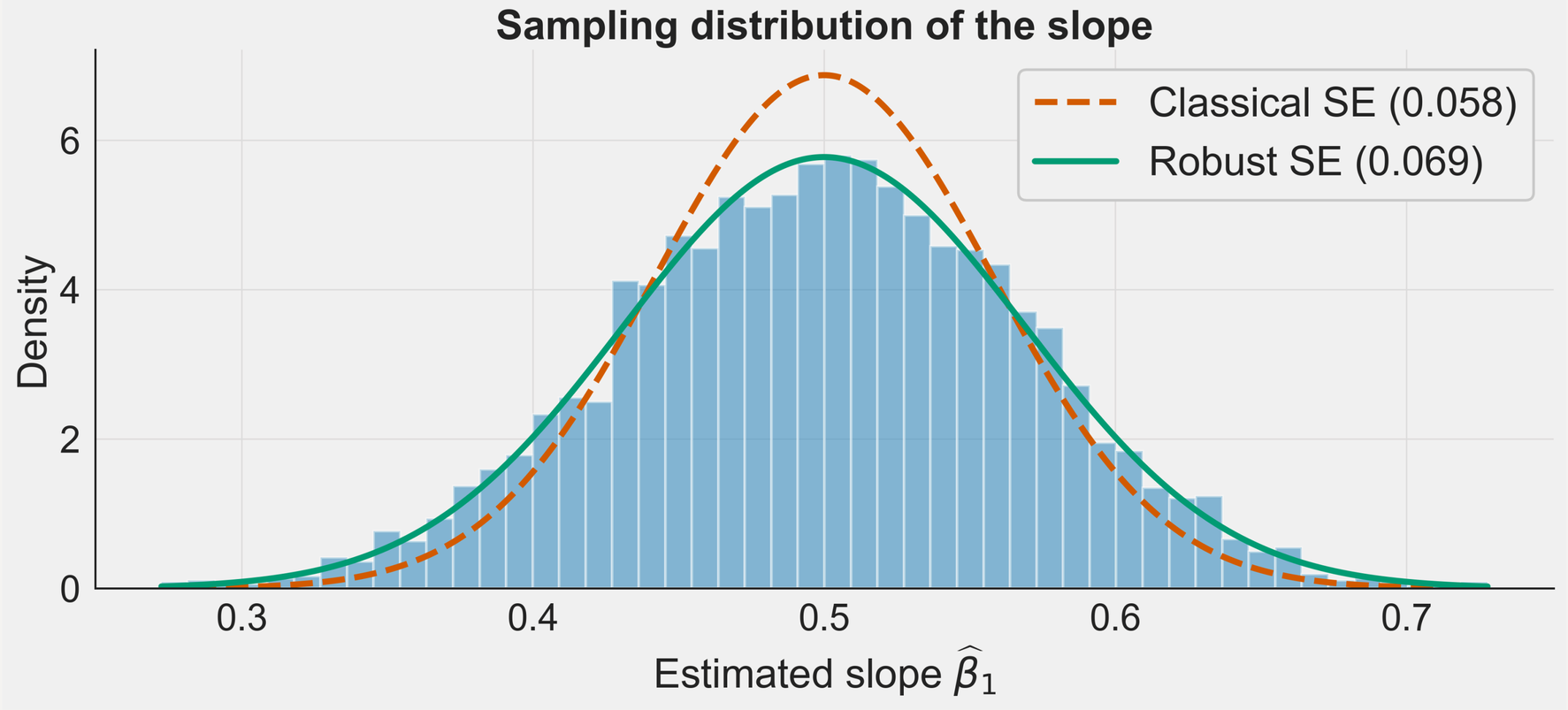
- The spread of the errors changes with $\mathbf{X}$
- *Are some observations noisier than others?*

# Why It Matters

- OLS is still unbiased
- OLS is no longer the best linear unbiased estimator
- The usual standard errors are wrong, in either direction
- The usual $t$-tests, $F$-tests, and confidence intervals have the wrong error rates, even in large samples

# The Standard Error Is Wrong

- 4,000 simulated samples, each with $n = 200$
- The true slope is 0.5, and the error variance grows with $x$
- Bars: the slope estimated in each sample
- Curves: the normal distribution implied by each standard error



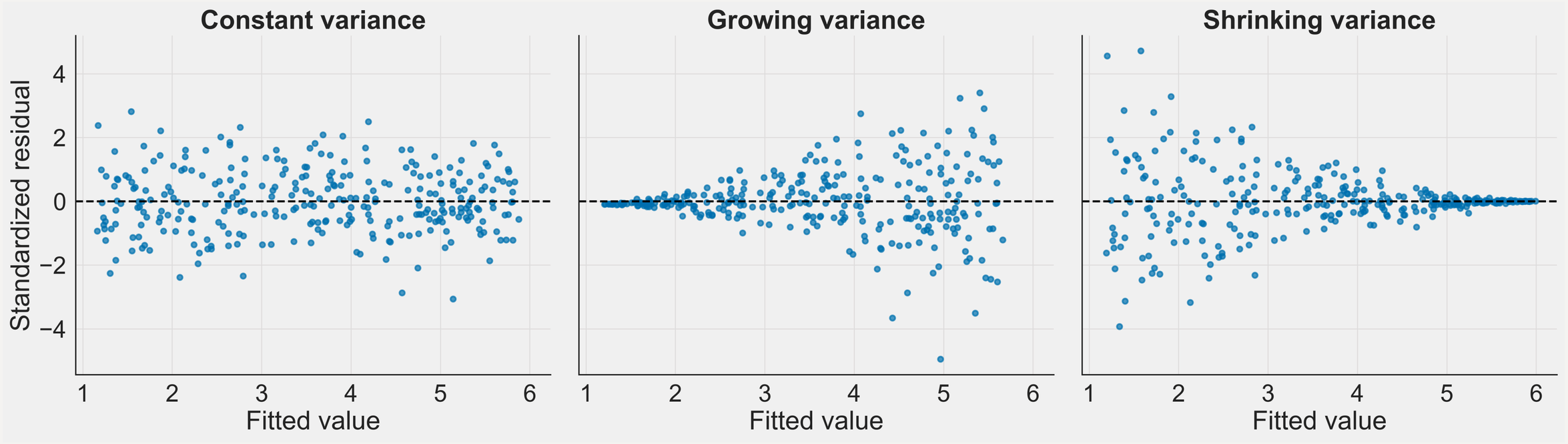
# What Causes It

- The outcome’s scale: spending, income, populations
- Groups with different variability
- Averaged data with different group sizes
- Binary outcomes in a linear model
- Misspecification, such as an omitted variable or the wrong functional form

# What It Implies

- Sometimes it is a feature of the data, and only the standard errors need fixing
- Example: attitudes on trade when one party’s voters are split into factions and the other’s are united
- Sometimes it is a symptom of misspecification
- Check the functional form and the outcome’s scale before patching the standard errors

# Constant vs. Changing Variance



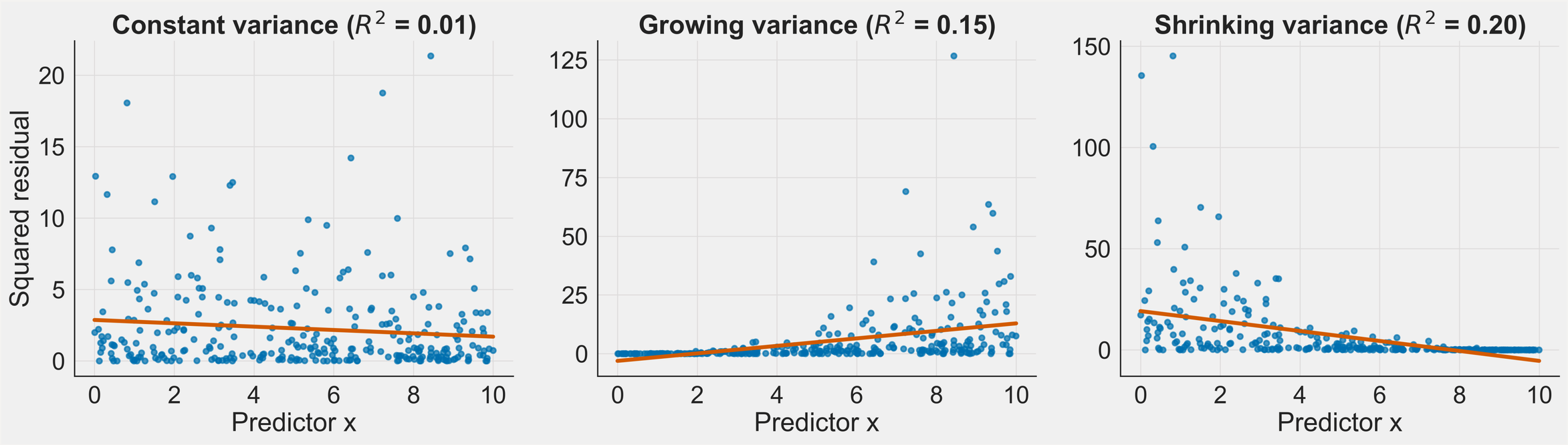
# A Test for Changing Spread

- A residual plot is a judgment call
- A gradual change in spread is hard to see
- *Is the spread of the residuals related to the predictors?*
- **Breusch-Pagan test**: a test of the null that the error variance is constant

# Squared Residuals Measure Spread

- The variance of an error is its expected squared value
- We see residuals, not errors, so $\widehat{\varepsilon}_i^2$ stands in for the variance at observation $i$
- Constant variance: the squared residuals are unrelated to the predictors
- Variance related to $x$: the squared residuals rise or fall with $x$

# Squared Residuals Against x



# The Breusch-Pagan Test

1. Fit the model and save the squared residuals $\hat{\varepsilon}_i^2$
2. Regress the squared residuals on the predictors
3. Compute $LM = nR^2$, where $n$ is the sample size and $R^2$ comes from step 2
4. Compare $LM$ to a $\chi^2$ distribution with $k$ degrees of freedom, where $k$ is the number of predictors

$$\hat{\varepsilon}_i^2 = \delta_0 + \delta_1 x_{1i} + \cdots + \delta_k x_{ki} + u_i, \qquad H_0 : \delta_1 = \cdots = \delta_k = 0$$

# Interpreting the Statistic

- Small $LM$: the predictors do not explain the squared residuals, so we fail to reject constant variance
- Large $LM$: they explain more than chance would, so we reject constant variance
- The direction of the relationship does not matter
- The cutoff is the $\chi^2_k$ critical value: 3.84 when $k = 1$ at the 5% level
- Failing to reject does not show that the variance is constant

# Results for the Simulated Data

| Case | $R^2$ | $LM = nR^2$ | $p$-value |
|---|---|---|---|
| Constant variance | 0.012 | 3.6 | 0.059 |
| Growing variance | 0.155 | 46.4 | $< 0.001$ |
| Shrinking variance | 0.202 | 60.6 | $< 0.001$ |

- $n = 300$ and $k = 1$, so the 5% cutoff is 3.84
- It only detects variance that moves linearly with the predictors

# Responding to Heteroscedasticity

- First, check the model: transform the outcome or add the missing structure
- If the unequal variance is a feature of the data, this creates two problems.

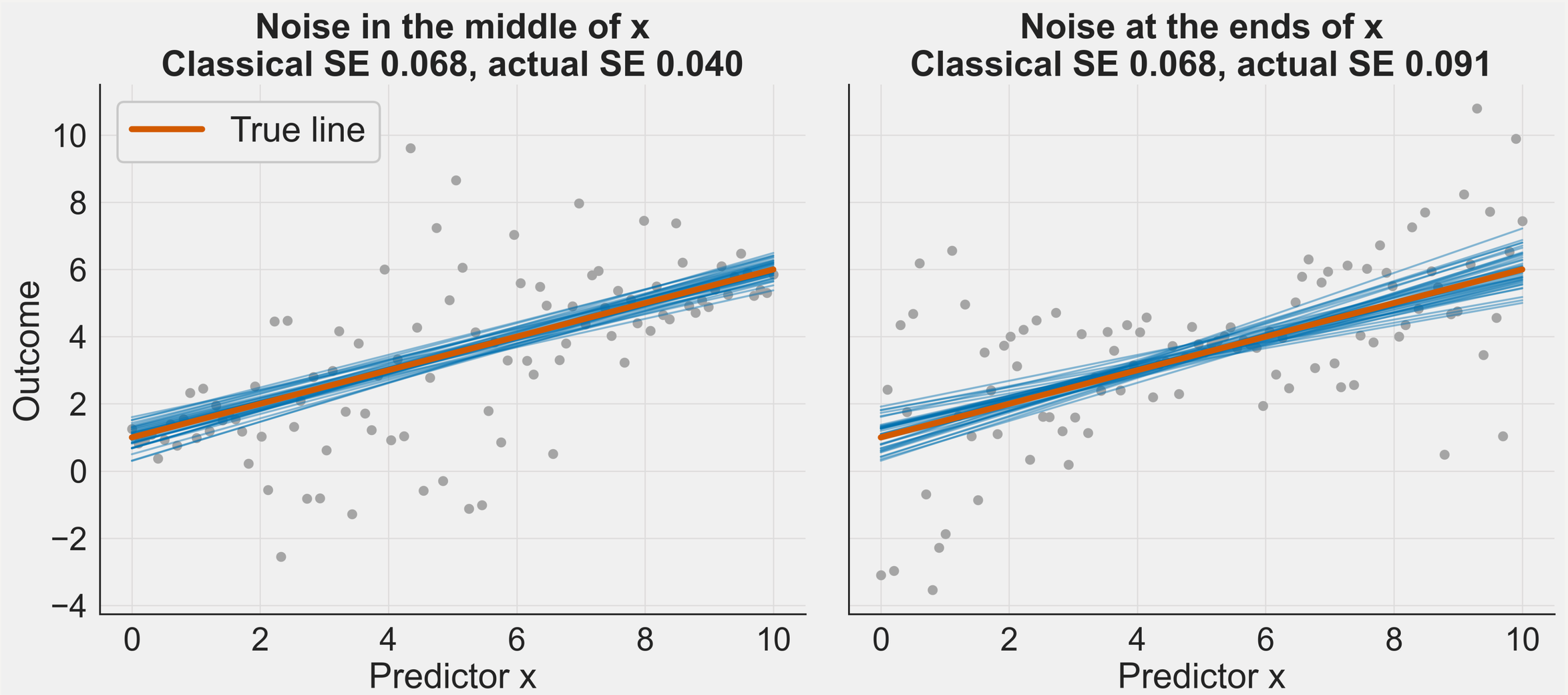
| Problem | Fix | Changes the coefficients? |
|---|---|---|
| Standard errors are wrong | Robust standard errors | No |
| OLS is no longer the best estimator | Weighted least squares | Yes |

# Robust Standard Errors

# Fixing the Standard Errors

- The OLS coefficients are still unbiased, so only the standard errors need fixing
- The usual formula pools all the residuals into one noise level, $\widehat{\sigma}^2$
- Pooling treats every observation as equally noisy
- *Does it matter which observations are noisy?*

# Where the Noise Sits Matters



# What Robust Standard Errors Do

- Keep the OLS coefficients
- Relax the assumption of a constant variance
- Measure each observation’s noise with its own squared residual
- Count that noise more for observations far from $\bar{x}$

# The Sandwich Formula

The variance of the OLS coefficients, with no assumption about the errors:

$$\text{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{X})^{-1} \; \mathbf{X}^\top \boldsymbol{\Omega} \mathbf{X} \; (\mathbf{X}^\top \mathbf{X})^{-1}$$

- Bread: depends only on $\mathbf{X}$
- Meat: where the errors enter
- $\boldsymbol{\Omega} = \text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X})$: an $n \times n$ matrix with the variance of each error on the diagonal and the covariance of each pair off it

# What Is in $\boldsymbol{\Omega}$?

Under homoscedasticity with uncorrelated errors:

$$\boldsymbol{\Omega} = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 \mathbf{I}$$

- Diagonal: $\operatorname{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma^2$, the same for every observation
- Off the diagonal: $\operatorname{Cov}(\varepsilon_i, \varepsilon_j \mid \mathbf{X}) = 0$

# Under Homoskedasticity

Substitute $\boldsymbol{\Omega} = \sigma^2 \mathbf{I}$. Since $\sigma^2$ is a number, it moves to the front:

$$(\mathbf{X}^\top \mathbf{X})^{-1}\ \mathbf{X}^\top \sigma^2 \mathbf{I}\, \mathbf{X}\, (\mathbf{X}^\top \mathbf{X})^{-1}\ = \sigma^2\, (\mathbf{X}^\top \mathbf{X})^{-1}\ \mathbf{X}^\top \mathbf{X}\, (\mathbf{X}^\top \mathbf{X})^{-1}$$

The meat cancels one slice of bread:

$$\mathrm{Var}(\,\widehat{\boldsymbol{\beta}}\ \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

- This only works because one $\sigma^2$ can be pulled out

# Allowing Different Variances

Now give each observation its own variance:

$$\boldsymbol{\Omega} = \begin{bmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n^2 \end{bmatrix}$$

No single $\sigma^2$ can be pulled out, so the meat is a sum over the rows of $\mathbf{X}$:

$$\mathbf{X}^\top \boldsymbol{\Omega} \mathbf{X} = \sum_{i=1}^{n} \sigma_i^2 \, \mathbf{x}_i \mathbf{x}_i^\top$$

# The Sandwich Estimator

- We cannot estimate each $\sigma_i^2$, since we see one draw per observation
- So we use $\hat{\varepsilon}_i^{\,2}$ in place of $\sigma_i^2$

$$\widehat{\mathrm{Var}}\left(\widehat{\boldsymbol{\beta}}\right) = (\mathbf{X}^\top \mathbf{X})^{-1} \left( \sum_{i=1}^{n} \hat{\varepsilon}_i^{\,2}\, \mathbf{x}_i \mathbf{x}_i^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1}$$

- Each term is noisy, but the noise averages out across the sum
- The result is a $(k+1) \times (k+1)$ matrix with each coefficient's estimated variance on the diagonal
- **Robust standard errors**: the square roots of those variances

# Versions of Robust Standard Errors

| Version | Replace $\hat{\varepsilon}_i^2$ with | Use for |
|---|---|---|
| HC0 | $\hat{\varepsilon}_i^2$ | Large samples |
| HC1 | $\hat{\varepsilon}_i^2 \cdot n/(n-k-1)$ | Stata |
| HC2 | $\hat{\varepsilon}_i^2/(1-h_{ii})$ | Experiments |
| HC3 | $\hat{\varepsilon}_i^2/(1-h_{ii})^2$ | Samples < ~250 |

- Residuals run smaller than errors, especially for unusual $x$
- $h_{ii}$: leverage, how unusual $\mathbf{x}_i$ is (next lecture)
- They agree in large samples, but software defaults differ

# Robust Standard Errors in Practice

- The coefficients do not change
- The standard errors can go up or down
- Valid without knowing the form of the heteroscedasticity
- They do not fix a wrong regression equation, such as one that omits a predictor
- The cost is some loss of precision compared with a correctly weighted estimator

# Clustered Errors

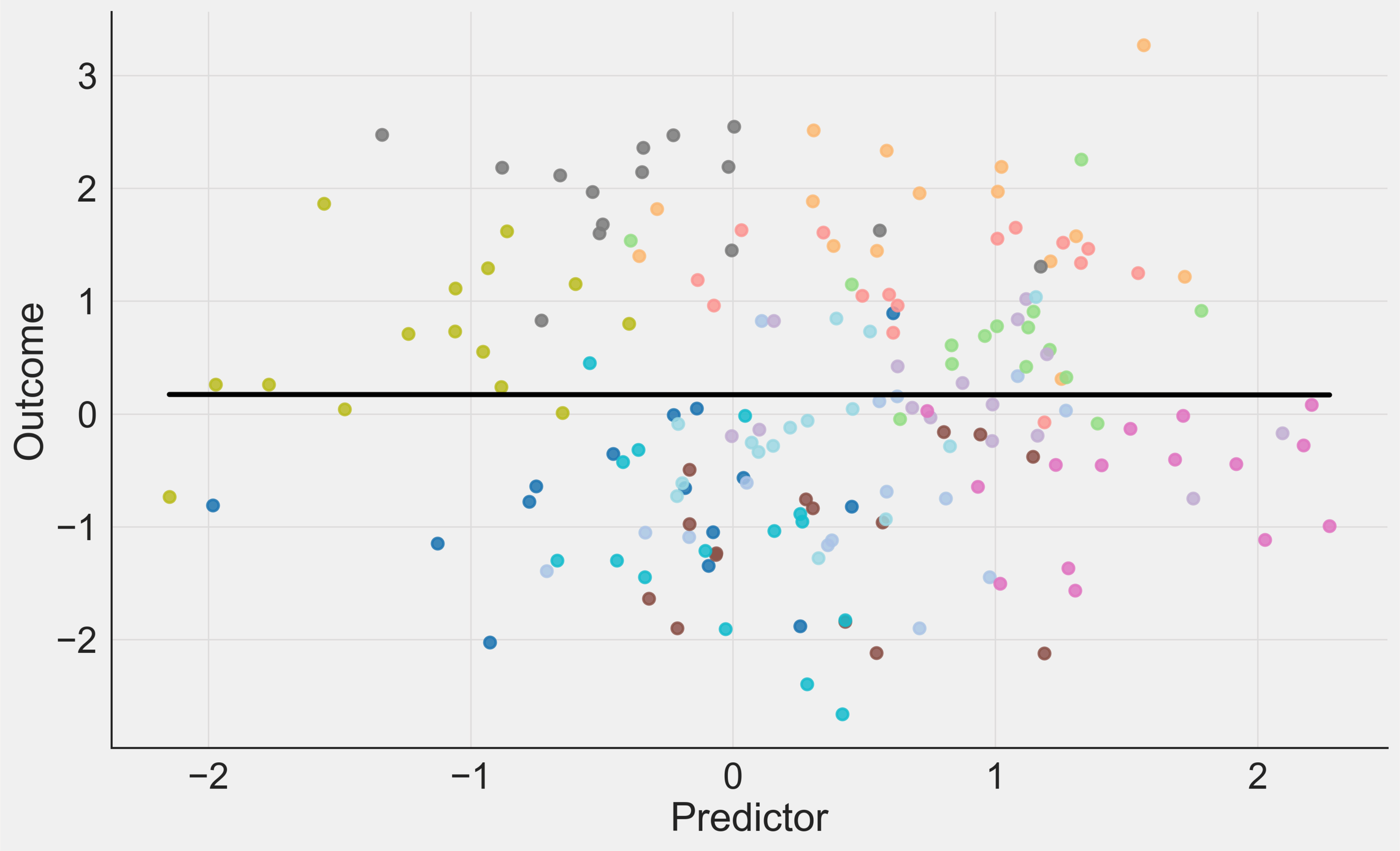
# The Problem

- **Cluster**: a group of observations that share something we did not model
- Examples: students in a school, legislators in a state, repeated observations of one unit
- Errors within a cluster are correlated
- *Does a second student from the same school tell us as much as a student from a new school?*

# Why It Matters

- OLS is still unbiased if the shared shocks are unrelated to the predictors
- The usual standard errors treat the observations as independent
- Correlated observations carry less information than independent ones
- The standard errors are too small

# Shared Shocks Within Clusters



# Cluster-Robust Standard Errors

$$\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \left( \sum_{g=1}^{G} \mathbf{X}_g^\top \widehat{\boldsymbol{\varepsilon}}_g \widehat{\boldsymbol{\varepsilon}}_g^\top \mathbf{X}_g \right) (\mathbf{X}^\top \mathbf{X})^{-1}$$

- $g = 1, \ldots, G$ indexes the clusters
- $\mathbf{X}_g$ and $\widehat{\boldsymbol{\varepsilon}}_g$: the rows of $\mathbf{X}$ and the residuals in cluster $g$
- Same bread, new meat: the sum runs over clusters, not observations

# Choosing the Clusters

- Cluster at the level where errors are correlated
- Cluster at the level of sampling or treatment assignment
- Need many clusters, roughly 30 to 50 or more
- The meat has one term per cluster, so $G$ acts as the sample size, not $n$
- With few clusters, standard errors are underestimated

# Weighted Least Squares

# Why Change the Estimator?

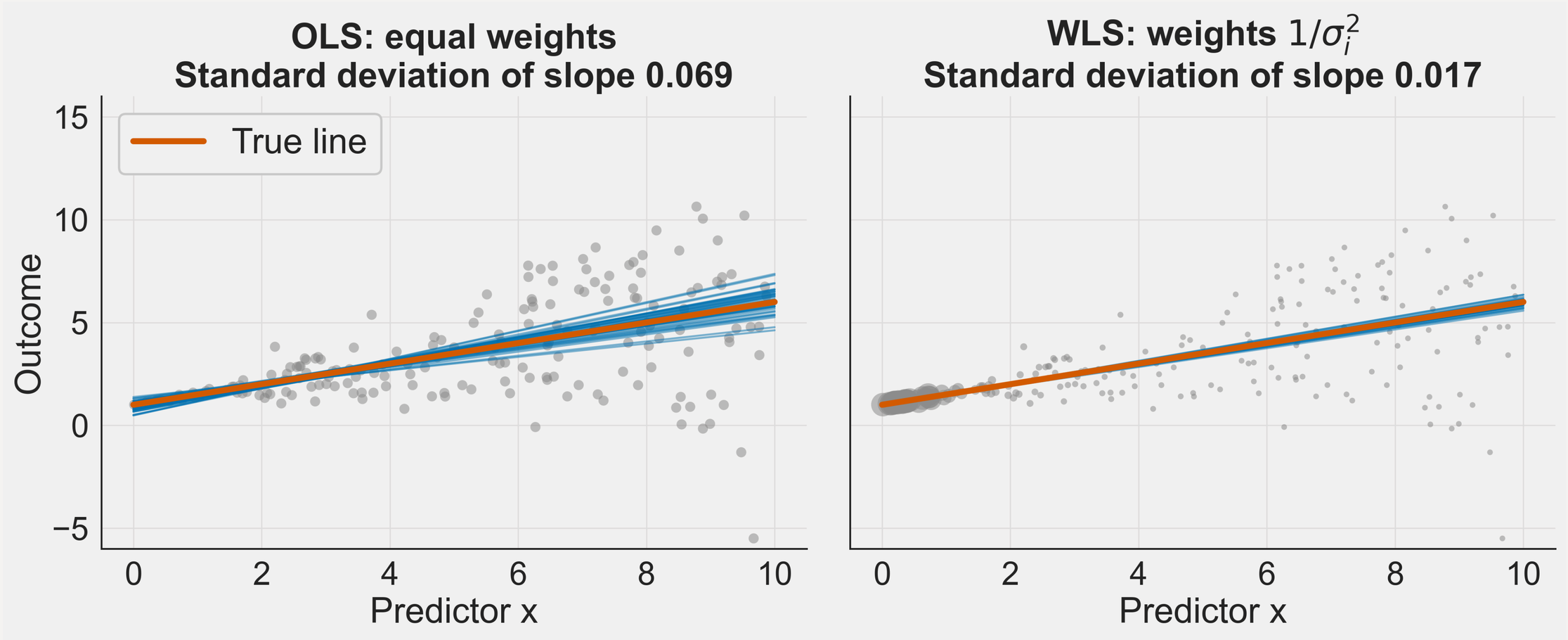
- Robust standard errors correct the standard errors, but the coefficients are still the OLS estimates
- OLS gives every observation the same weight, whatever its error variance
- Under heteroscedasticity, OLS is unbiased but not BLUE: another linear unbiased estimator has a smaller variance
- Example: a state average from 5,000 respondents has 1/100 the variance of one from 50
- *Should those two observations have the same weight?*

# What WLS Does

- **Weighted least squares (WLS)**: OLS with a weight on each observation
- Precise observations, those with a small error variance, receive large weights
- Imprecise observations, those with a large error variance, receive small weights
- The fitted line therefore stays closer to the precise observations
- The coefficients change, and they vary less from sample to sample

# WLS Reduces Sampling Variability

- Each blue line is one sample's fit; larger points have more weight
- Both center on the true line, but the WLS lines vary far less



# The WLS Objective

WLS multiplies each squared residual by a weight $w_i$:

*[Math Processing Error]*

- The weight is the inverse of the error variance: $w_i = 1/\sigma_i^2$
- A large $w_i$ makes a large residual at observation $i$ more costly
- If every $\sigma_i^2$ is equal, every weight is equal and WLS is OLS

# Why Inverse-Variance Weights?

Start with the model and divide every term by $\sigma_i$:

$$\frac{y_i}{\sigma_i} = \frac{\mathbf{x}_i^\top}{\sigma_i}\boldsymbol{\beta} + \frac{\varepsilon_i}{\sigma_i}$$

- $\boldsymbol{\beta}$ does not change, because both sides are divided by the same number
- The new error has the same variance for every observation:

$$\mathrm{Var}(\varepsilon_i/\sigma_i \mid \mathbf{X}) = \sigma_i^2/\sigma_i^2 = 1$$

- With uncorrelated errors, the rescaled model meets the Gauss-Markov assumptions, so OLS on it is BLUE

# OLS on the Rescaled Model Is WLS

OLS on the rescaled model minimizes the sum of squared rescaled residuals:

*[Math Processing Error]*

- The right side is the WLS objective, with $w_i = 1/\sigma_i^2$
- So WLS with these weights is BLUE
- Weights that are not proportional to $1/\sigma_i^2$ cannot give a smaller variance

# The WLS Estimator

Minimizing the WLS objective gives:

*[Math Processing Error]*

- This is the OLS formula with $\mathbf{W}$ inserted; if $\mathbf{W} = \mathbf{I}$, it is OLS
- With $w_i = 1/\sigma_i^2$, $\mathbf{W} = \mathbf{\Omega}^{-1}$ and the variance simplifies:

$$(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{\Omega} \mathbf{W} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$$

$$\text{Var}(\widehat{\boldsymbol{\beta}}_{WLS} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$$

# Generalized Least Squares

- WLS assumes uncorrelated errors, so $\boldsymbol{\Omega}$ is diagonal
- **Generalized least squares (GLS)**: the extension of WLS to any known $\boldsymbol{\Omega}$, including correlated errors

$$\widehat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}^\top \boldsymbol{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Omega}^{-1} \mathbf{Y}$$

- If $\boldsymbol{\Omega}$ is diagonal, $\boldsymbol{\Omega}^{-1} = \mathbf{W}$ and GLS is WLS
- GLS cannot be computed unless we know $\boldsymbol{\Omega}$, and we usually do not
- **Feasible GLS**: GLS with an estimate of $\boldsymbol{\Omega}$ in place of $\boldsymbol{\Omega}$

# A Model for the Variances

- With a diagonal $\boldsymbol{\Omega}$, feasible GLS is WLS with estimated weights
- Estimating the weights requires a second model, alongside the regression
- **Mean model**: the regression equation, $E[y_i \mid \mathbf{X}] = \mathbf{x}_i^\top \boldsymbol{\beta}$
- **Variance model**: an equation for how $\sigma_i^2$ depends on the predictors, such as:

$$\sigma_i^2 = \exp(\gamma_0 + \gamma_1 x_{1i} + \cdots + \gamma_k x_{ki})$$

- The exponential makes every $\sigma_i^2$ positive, whatever the values of the $\gamma$s

# Feasible GLS in Four Steps

1. Fit OLS and save the residuals $\hat{\varepsilon}_i$
2. Regress $\ln(\hat{\varepsilon}_i^2)$ on the predictors and save the fitted values $\hat{g}_i$
3. Compute the estimated variances, $\hat{\sigma}_i^2 = \exp(\hat{g}_i)$
4. Fit WLS with the weights $w_i = 1/\hat{\sigma}_i^2$

*Why the log of the squared residuals?*

A regression of $\hat{\varepsilon}_i^2$ on the predictors can have negative fitted values, which cannot be variances. Fitted values from the log regression can be negative too, but $\exp(\hat{g}_i)$ is always positive.

# When the Weights Are Wrong

- Weights are wrong if they are not proportional to $1/\sigma_i^2$, as when the variance model is wrong
- WLS is still unbiased if the mean model is correct, meaning $E[\varepsilon_i \mid \mathbf{X}] = 0$
- But it is not BLUE, and its variance can be larger than the variance of OLS
- Also, $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ is the wrong variance, so the usual WLS standard errors are wrong
- Robust standard errors for WLS correct this; always use them with feasible GLS

# Linear Models Are Approximations

- Unbiasedness and BLUE assume that $E[y_i \mid \mathbf{X}]$ is exactly linear, which is rarely true
- Otherwise, each estimator gives a linear approximation to the true mean
- OLS counts all observations equally; WLS fits best where the weights are large
- Example: if the true mean is $6\sqrt{x_i}$ and the variance grows with $x_i$, the OLS slope approaches 1.5 and the WLS slope approaches 2.3
- So a large gap between OLS and WLS estimates is a reason to check the mean model

# Robust SEs or WLS?

| What we know about $\sigma_i^2$ | Estimator | SEs |
|---|---|---|
| Known up to a constant | WLS | Usual WLS |
| Unknown, variance model fitted | Feasible GLS | Robust |
| Unknown, no variance model | OLS | Robust |

- OLS with robust SEs is the default, because it requires no variance model
- Its cost is less precision than WLS with correct weights
- Neither corrects a wrong mean model, such as an omitted predictor

# Putting It Together

# Diagnostics at a Glance

| Problem | Tools | Responses |
|---|---|---|
| Non-normal residuals | Histogram, QQ plot | Transform, fix the model |
| Heteroscedasticity | Residual plot, Breusch-Pagan | Robust SEs, WLS |
| Clustering | Study design | Clustered SEs |

# Summary

- Diagnostics look for structure in the residuals, which points to a failed assumption
- Non-normal residuals do not bias OLS, but they often signal a misspecified model
- Heteroscedasticity leaves OLS unbiased but makes the usual standard errors wrong
- Clustered errors leave OLS unbiased but make the usual standard errors too small

# Key Definitions

# Key Definitions

- **Residual plot**: residuals against fitted values, used to spot curvature, changing spread, and unusual points
- **Prediction interval**: an interval for a single new outcome, which depends on the error distribution
- **Quantile**: the value $q_\tau$ below which a share $\tau$ of the distribution falls
- **QQ plot**: residual quantiles against normal quantiles; points on the line mean normality

# Key Definitions

- **Shapiro-Wilk test**: a test of the null that the residuals are normal; $W$ near 1 means normal
- **Skewness**: asymmetry of a distribution; 0 for a normal
- **Excess kurtosis**: tail weight relative to a normal; 0 for a normal
- **Jarque-Bera test**: a test of normality based on skewness and excess kurtosis; $JB \approx \chi^2_2$ under the null

# Key Definitions

- **Heteroscedasticity**: error variance that differs across observations
- **Breusch-Pagan test**: a test of constant variance; regress the squared residuals on the predictors
- **LM statistic**: $LM = nR^2$ from that regression; approximately $\chi^2_k$ under the null
- **Robust standard errors**: standard errors that stay valid under heteroscedasticity (the sandwich estimator)
- **Sandwich estimator**: *[Math Processing Error]*

# Key Definitions

- **Cluster**: a group of observations that share something we did not model
- **Clustered errors**: errors correlated within groups of observations
- **Cluster-robust standard errors**: a sandwich estimator that sums over clusters instead of observations

# Key Definitions

- **Weighted least squares**: minimizes $\sum w_i(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$ with $w_i = 1/\sigma_i^2$
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